

Jacob Democko

jacob.democko@gmail.com | (954) 547-0511 | Berkeley, CA | [Portfolio](#)

WORK EXPERIENCE

Lawrence Livermore National Lab

Aug. 2023 – Present

Senior Mechanical Engineer

Livermore, CA

- LLNL is a premier \$3B+ federally funded research and development center (FFRDC) employing over 9,000 multidisciplinary experts dedicated to solving the nation's most complex nuclear security and energy challenges.
 - As a senior mechanical engineer in Precision Systems & Manufacturing, I work on a variety of cutting-edge projects in the fields of precision design, metrology, micro- and nanofabrication, micro-electrical mechanical systems (MEMS), and tooling and process improvement.
 - **Key Results:** Delivered a mirror-based metrology fixture and inspection process for OGP systems, enabling sub-10 μm measurement of previously inaccessible geometries and improving inspection throughput for high-precision components for fusion energy research at the National Ignition Facility.
- Advanced design, fabrication, fixturing and design documentation for a 3D brain-on-chip device, moving from BSL-2 to BSL-3 high containment.

Lucid Motors

May. 2021 – Aug. 2023

Product Design Engineer

Newark, CA

- Lucid Motors is a pioneering luxury electric vehicle startup with a \$2.5B market cap, 6500 employees, and achieved production of the Air Sedan and Lucid Gravity SUV, delivering advanced automotive technology.
- As a production design engineer, I led the development of key thermal system components (radiator, condenser, fan system, coolant system) and had expansive ownership over global supplier management, team coordination, and design verification and documentation.
 - **Key Results:** Achieved 50% reduction in unit cost for front-end cooling module through design simplification, reduced manufacturing cost through achieving BOM optimization and subassembly. Verified airflow targets exceeded through full-scale wind tunnel testing.

RMF Engineering

Jan. 2019 – Aug. 2020

Mechanical Infrastructure Design Consultant

Raleigh, NC

- Executed 20+ mechanical infrastructure projects from kickoff through construction administration and close-out
- Led mechanical layout and utility plant planning for new construction and retrofit projects, supporting system sizing for chilled water, high-pressure steam systems, pumps, and chillers.

EDUCATION

University of California, Davis & Berkeley

Davis & Berkeley, CA

M.S., Mechanical Engineering (Davis), Graduate Certificate in Technology Policy (Berkeley), Defense & Dual Use Technology Club Coursework: Advanced Energy Systems, Micro & Nano Biotechnology, Sustainable Manufacturing, Technology & Nuclear Policy

- **GPA: 4.0/4.0**

North Carolina State University

Raleigh, NC

B.S., Mechanical Engineering, graduated Summa Cum Laude

- **GPA: 3.89/4.0**

CERTIFICATIONS, SKILLS & INTERESTS

- **Certifications:** DOE Q Security Clearance; Engineer in Training (EIT)
- **Technologies:** CAD (Catia, Creo); CAE; Data Analysis (MATLAB, Python); KLayout; NI LabVIEW; Jira
- **Skills:** Communication; DFM/DFA; GD&T; Tolerance and Repeatability Analysis; Positive Attitude
- **Interests:** Fitness (Weightlifting, Running); Traveling; Restaurants, Automotive (Cars & Motorcycles); Comedy